

Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection

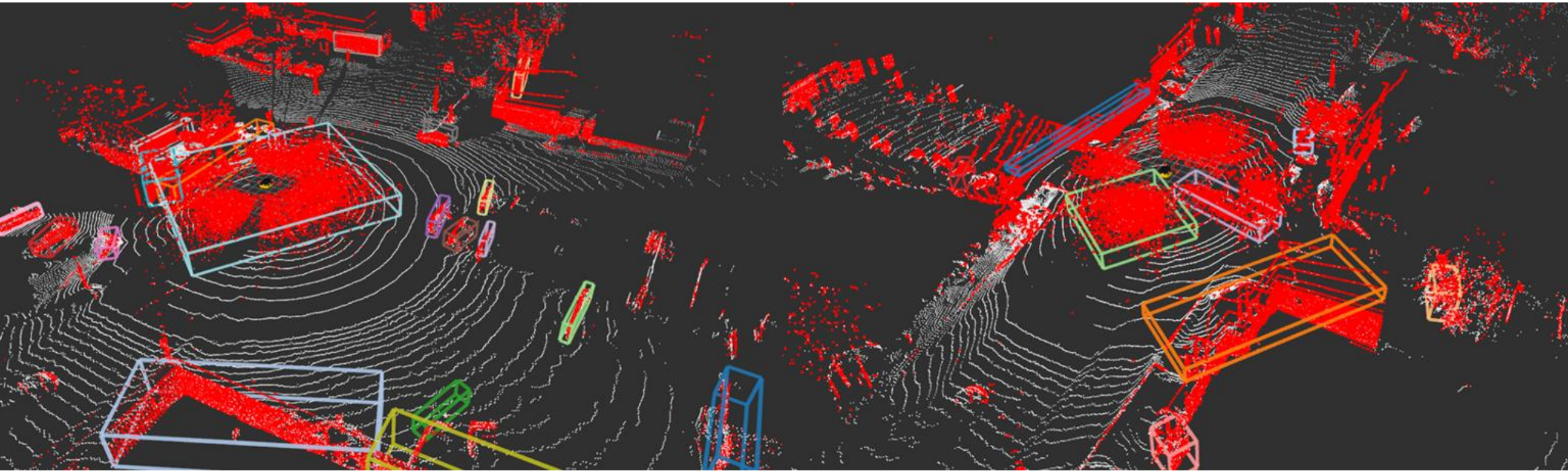
PhD candidate: Yiming Yang

Advisor: Dr. Jeremy P. Bos



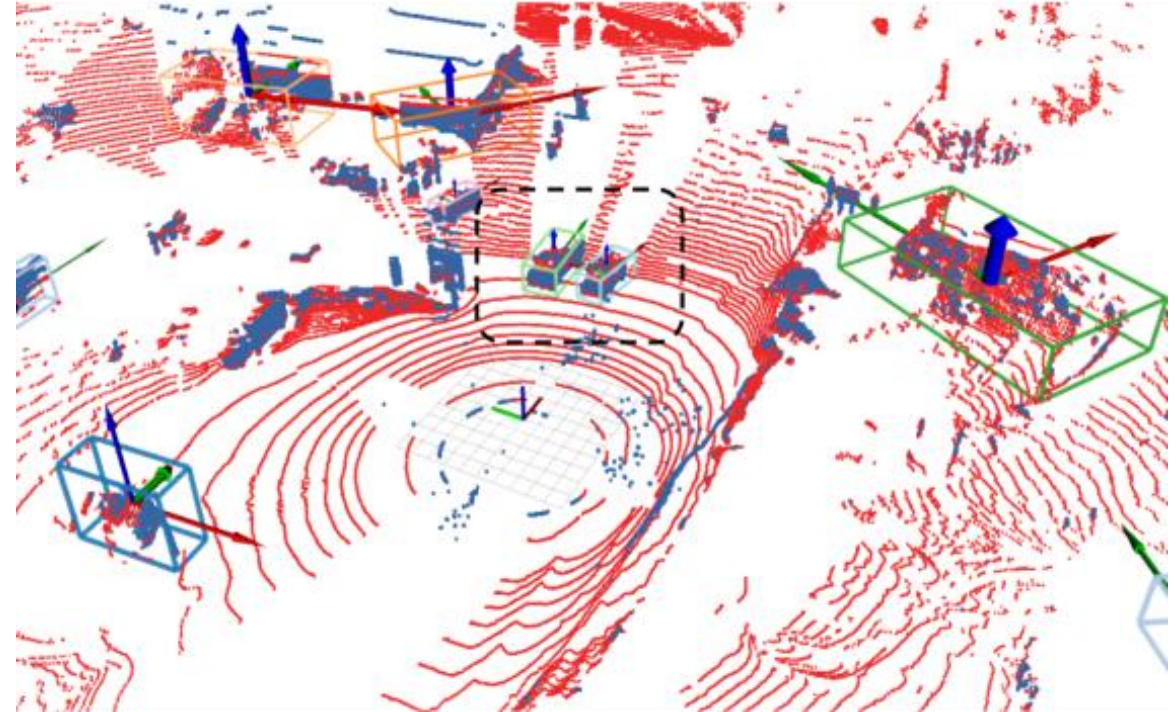
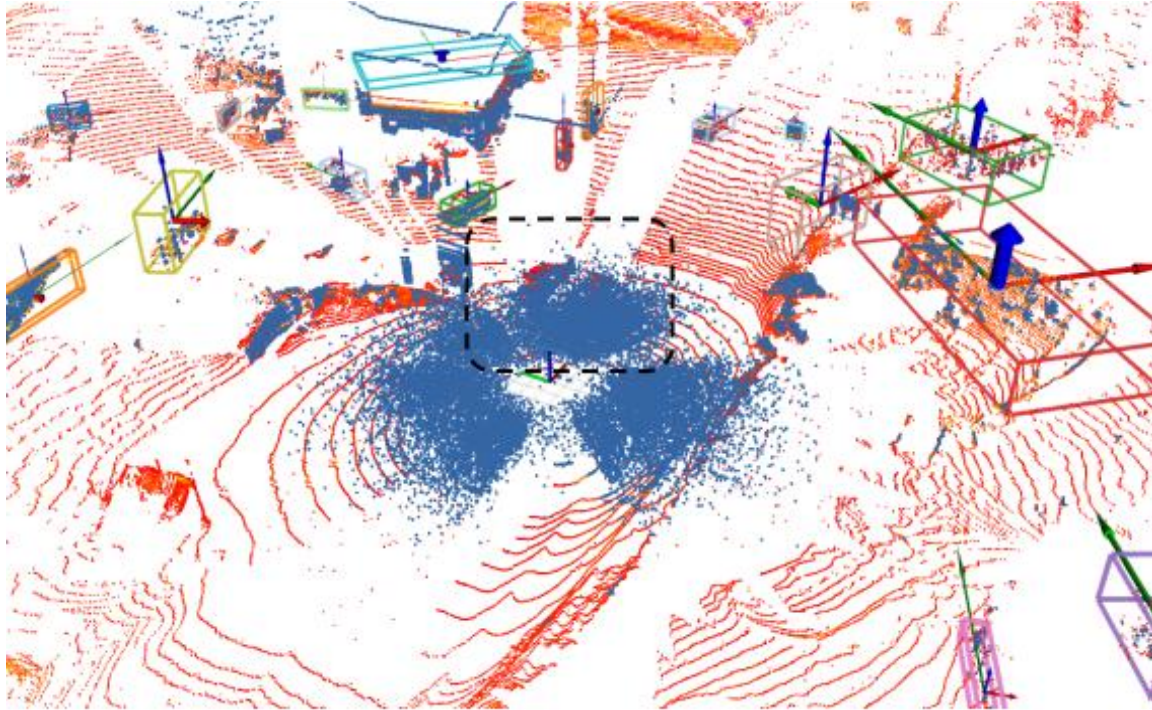
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Impact of Snow on LiDAR Perception - Backscatter



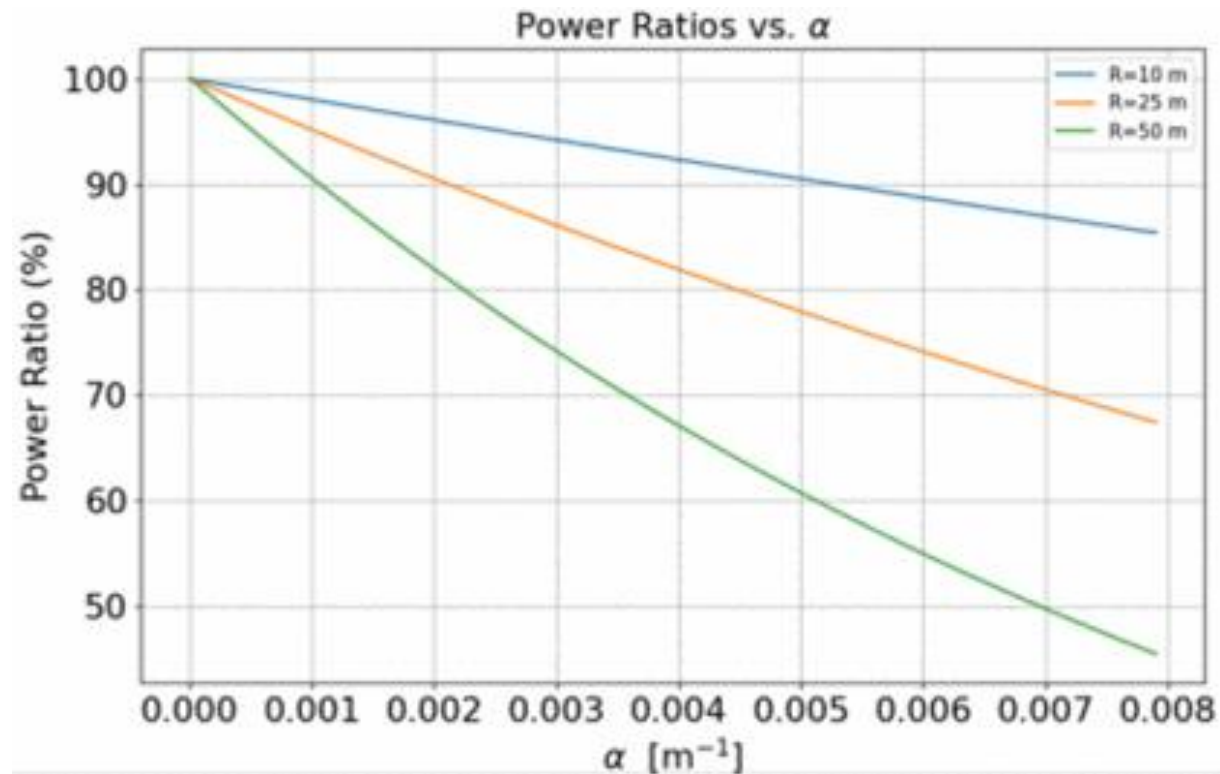
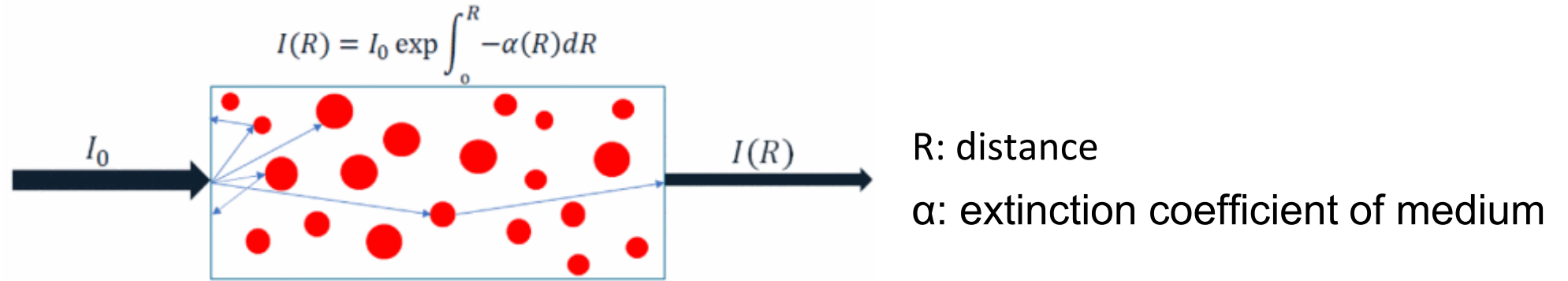
[JP Bos et al., 2021]

Impact of Snow on LiDAR Perception - Occlusion



[A Kurup et al., 2021]

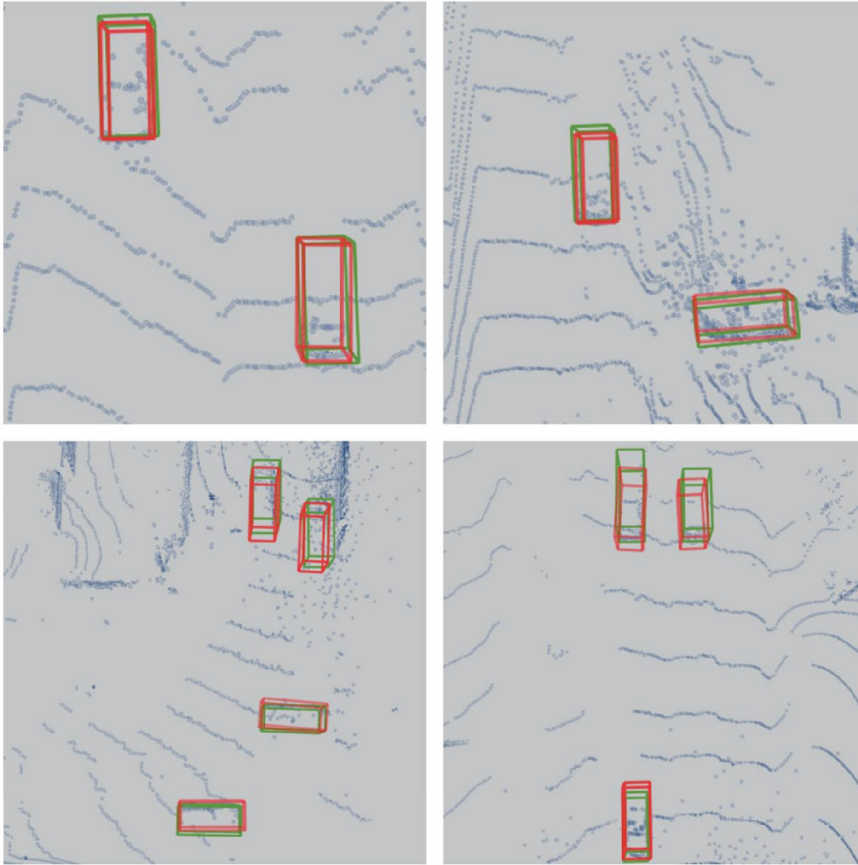
Impact of Snow on LiDAR Perception – 3. Attenuation



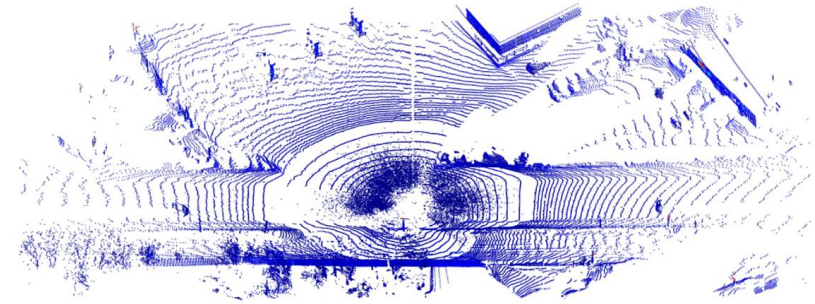
[Farhani et al., 2024]

Roadmap - LiDAR in Snowy Environments

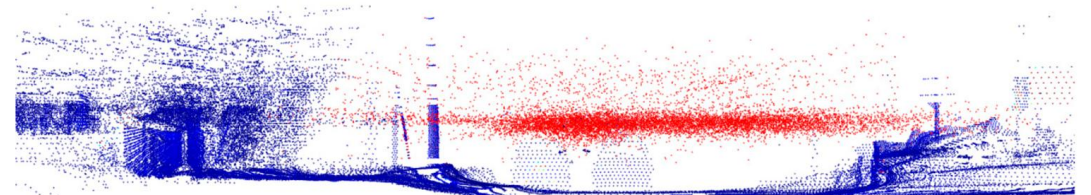
- Characterizing Impact (Problem)
- Mitigation Strategies (Solution)



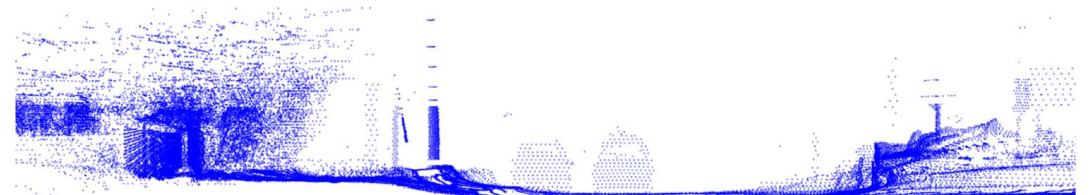
[Fan et al., 2024]



(a)



(b)

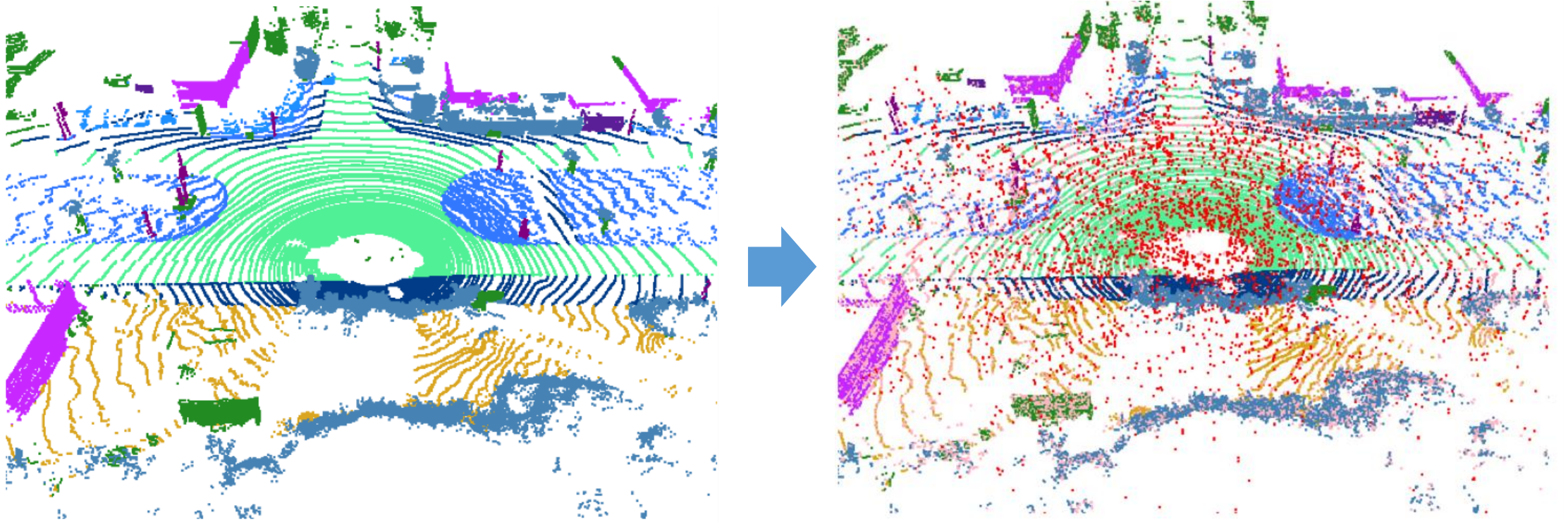


(c)

[Le et al., 2023]

Characterizing Impact - Existing research

- Physically-based Snow simulator
 - KITTI dataset
 - Models airborne snow physics
 - Accumulated snow is out of scope



[Hahner et al., CVPR 2022, Zhao et al. IEEE TIP 2025]

Characterizing Impact - Existing research

- Evaluation on real-world snowy datasets
 - Cross-Domain Variation
 - Coarse category (light vs. heavy)

Datasets	LiDAR Attribute		Annotation		Frames
	Models	Beams	Snow	Targets	
Boreas	Velodyne Alpha-Prime	128	✓	✗	1000
CADC	Velodyne VLP-32C	32	✓	✓	500
RADIATE	Velodyne HDL-32E	32	✓	✗	500
WADS	—	64	✓	✗	1000

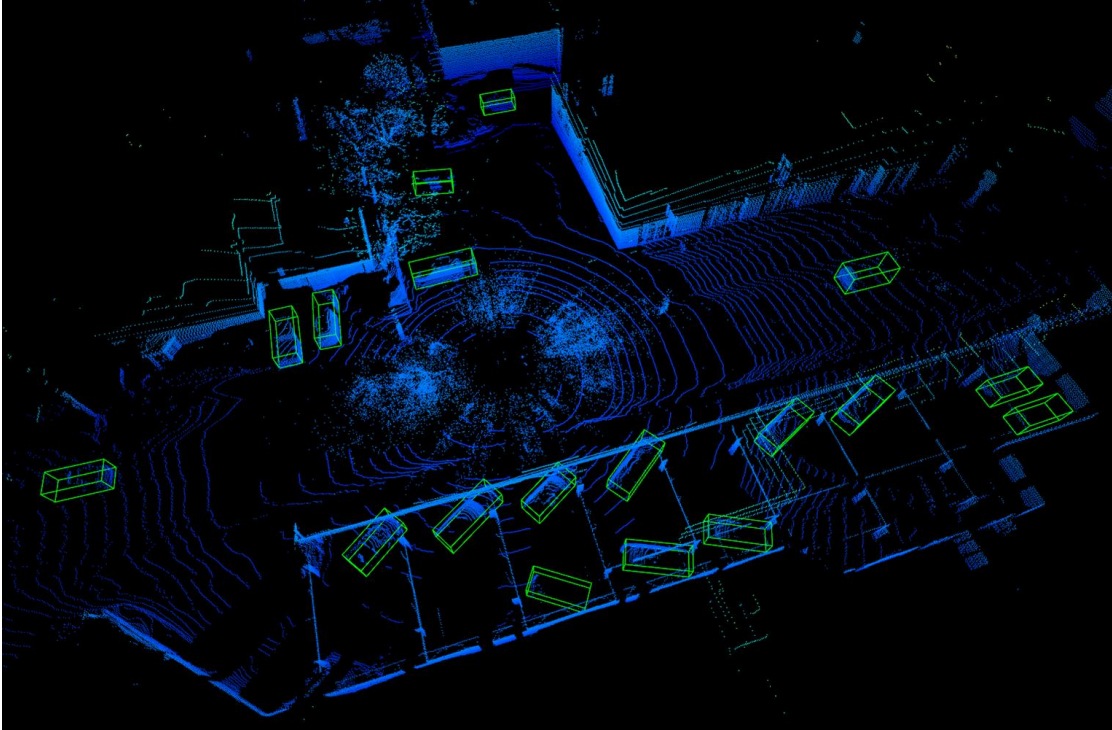
[Sun et al., ITS 2024]

Research Gap

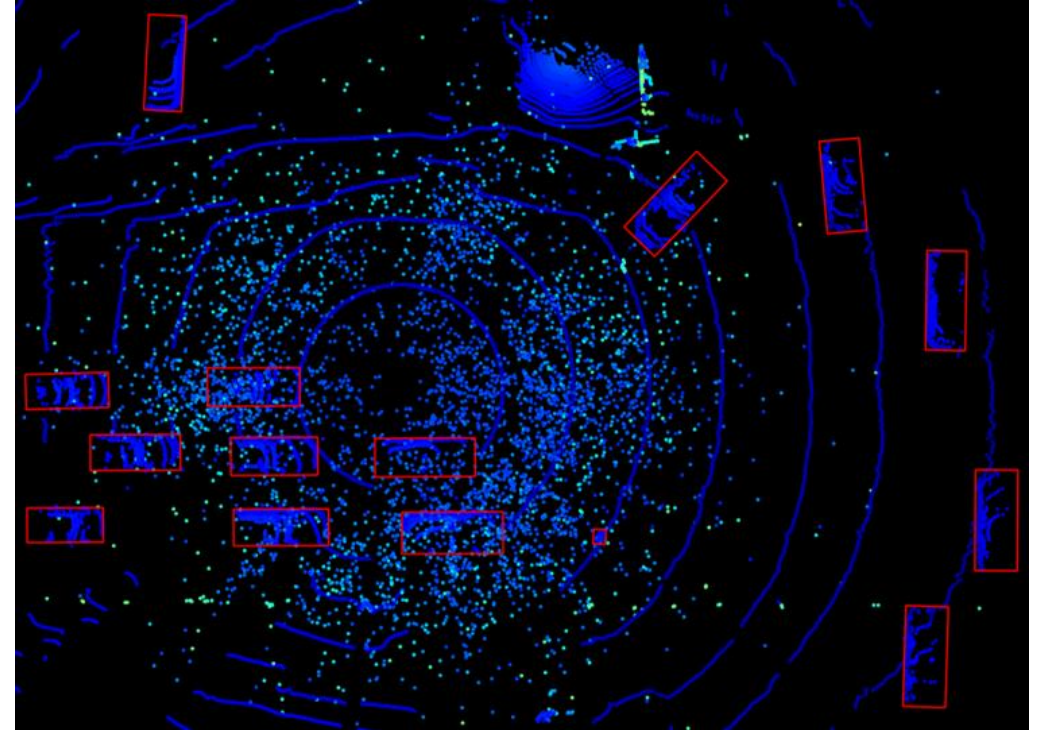
- Limitations in Current Snow Classification
 - Reliance on coarse weather categories
- Simulation-based
 - Simulate air-borne noise while ignoring surface accumulation
- Real-world Data
 - Cross-Domain Variation

Proposed Approach

- Two Real-world large datasets, Conduct Evaluations Separately



WADS-3D:
2,050 labeled frames
(4,100 KITTI standard frames)



CADC [Pitropov et al., 2020]:
7,000 labeled frames
(14,000 KITTI standard frames)

WADS: Winter Adverse Driving dataSet

- Multi-modal
 - Lidar
 - RGB
 - Thermal
 - GNSS
- Extreme-snow
- Large-scale
 - > 45 TB

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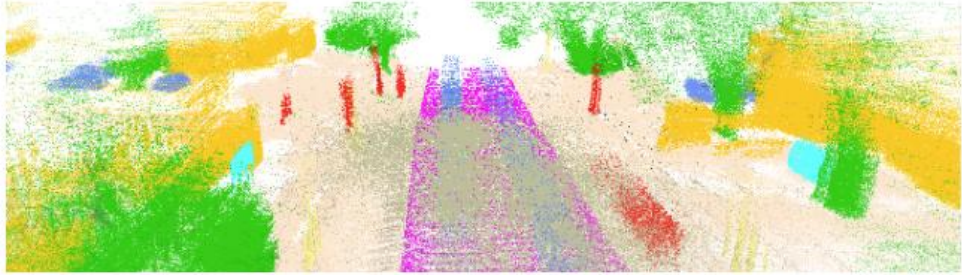
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WINTER ADVERSE DRIVING DATASET

Collected in the snow belt region of Michigan's Upper Peninsula, The Winter Adverse Driving dataset (WADS) is the first multi-modal dataset featuring dense point-wise labeled sequential LiDAR scans collected in severe winter weather. Over 26 TB of multi-modal data has been collected of which over 7 GB of LiDAR point clouds (3.6 billion points) have been labeled (semanticKITTI format) and made available here. This outdoor dataset introduces falling snow and accumulated snow along with all the semanticKITTI classes. The researchers believe this dataset will further AV tasks like semantic and panoptic segmentation, object detection and tracking, and localization and mapping in conditions of moderate to severe snow.

Recommended Citation

Kurup, A., & Bos, J. (2021). The Winter Adverse Driving dataset (WADS). Retrieved from: <https://digitalcommons.mtu.edu/wads/>

WADS-3D: Object detection under extreme snow

- Based on WADS
- Large scale
 - 30,104 labeled cars
- Complex Urban Env
 - High-clutter Env
(Downtown
Houghton &
Hancock)

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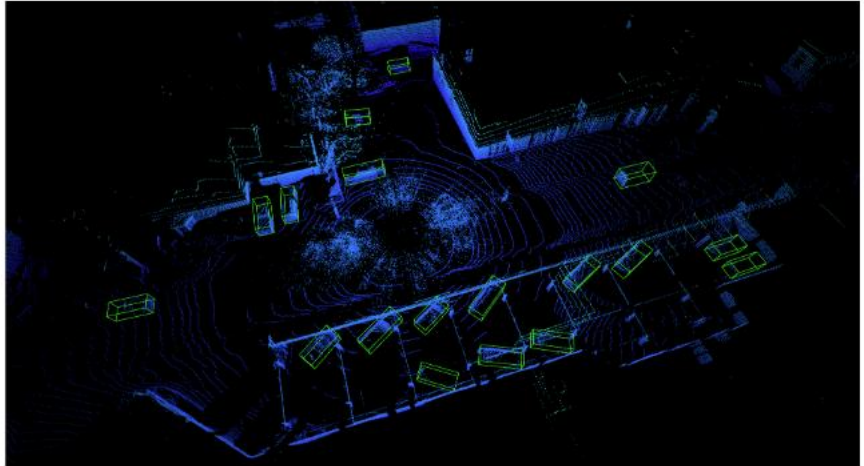
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WADS-3D

WADS-3D is a critical supplementary dataset providing 3D bounding box labels specifically for use with the existing Lidar point cloud frames from the WADS (Winter Adverse Driving Dataset). The dataset's core purpose is to facilitate the quantitative analysis of how different harsh winter conditions, including varying snow fall intensities and winter storms, influence the performance of Lidar-based 3D object detection algorithms. WADS-3D includes a total of 22 individual 3D bounding box label files: 21 files correspond to sequences already available in the original WADS dataset, and one file covers a newly released Lidar frame sequence (Sequence 77), thereby expanding the WADS coverage. Users must download the corresponding raw Lidar frames from the original WADS dataset repository (<https://digitalcommons.mtu.edu/wads/>) to utilize these labels.

Snow metric and Detectors

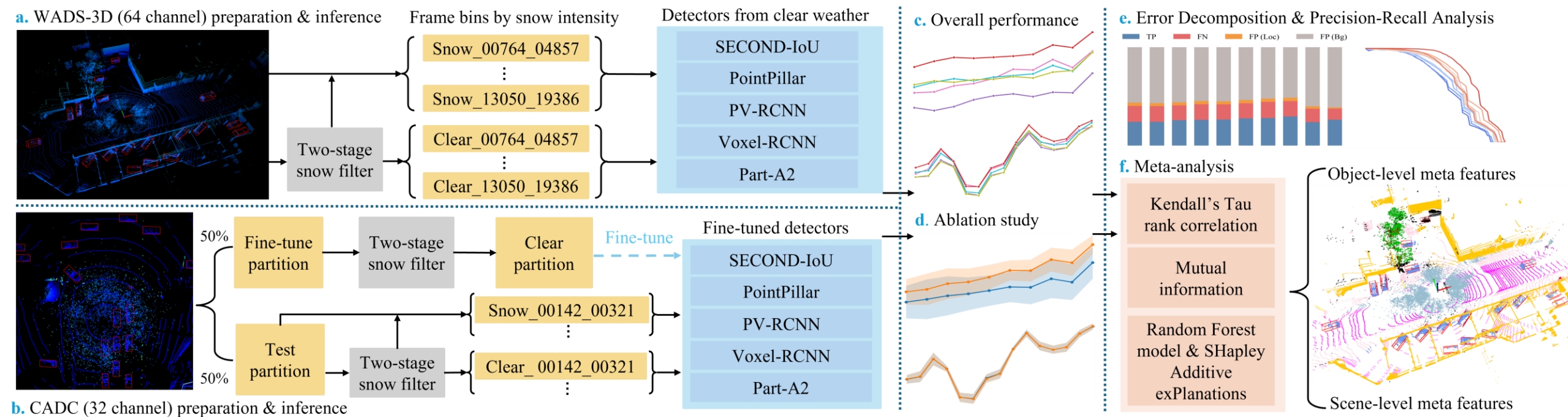
Granular Snow metric : Number of raw snow points

Neural detectors:

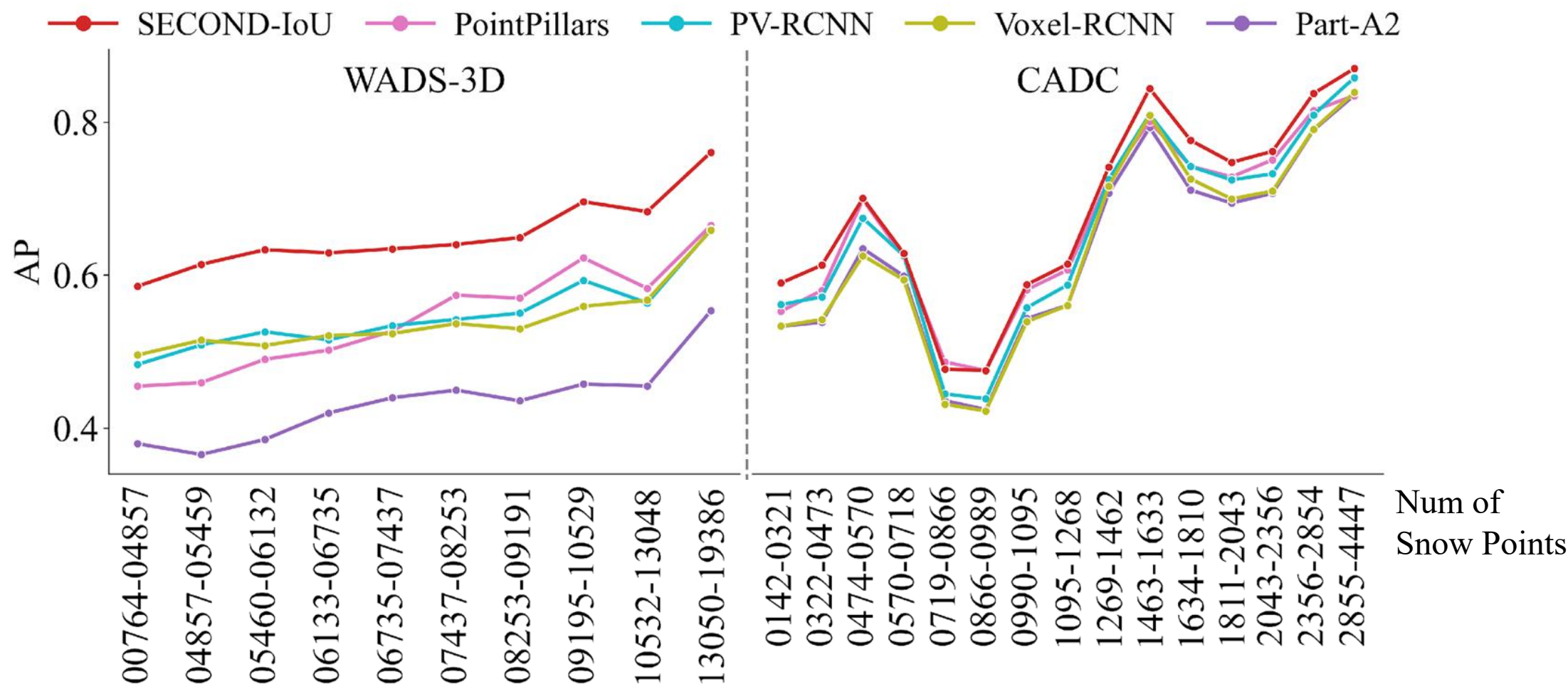
Detector	Architecture	Stages
SECOND-IOU	Voxel-based	One
Voxel R-CNN	Voxel-based	Two
PointPillar	Pillar-base	One
SECOND	Point-based	Two
PV-RCNN	Hybrid	Two

[OpenPCDet, 2020]

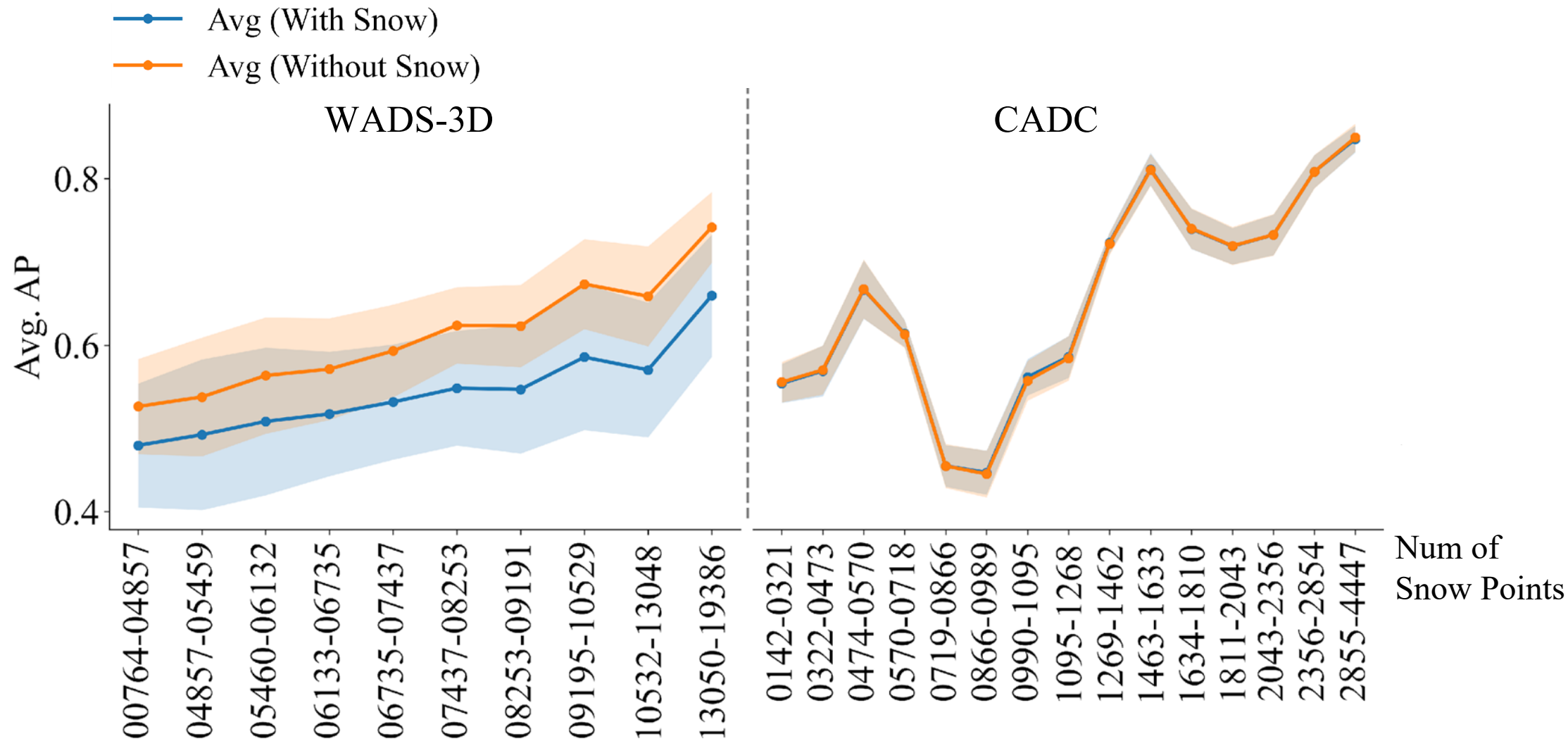
Evaluation Pipeline



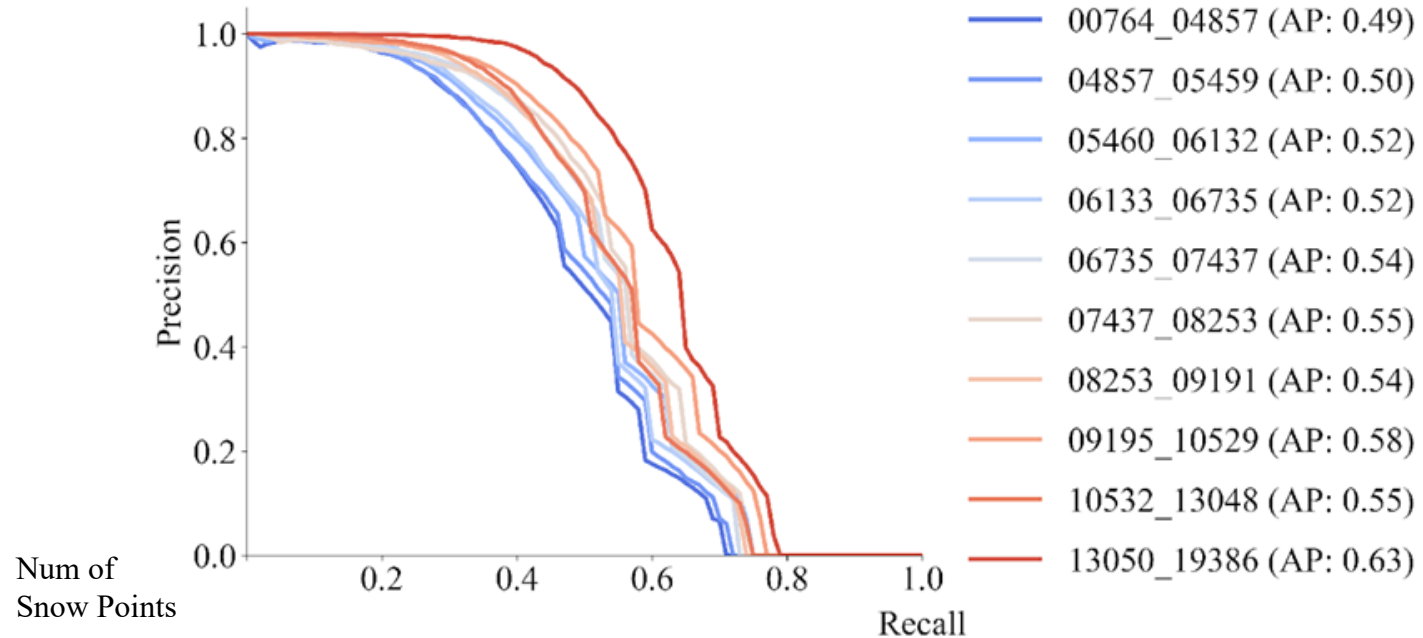
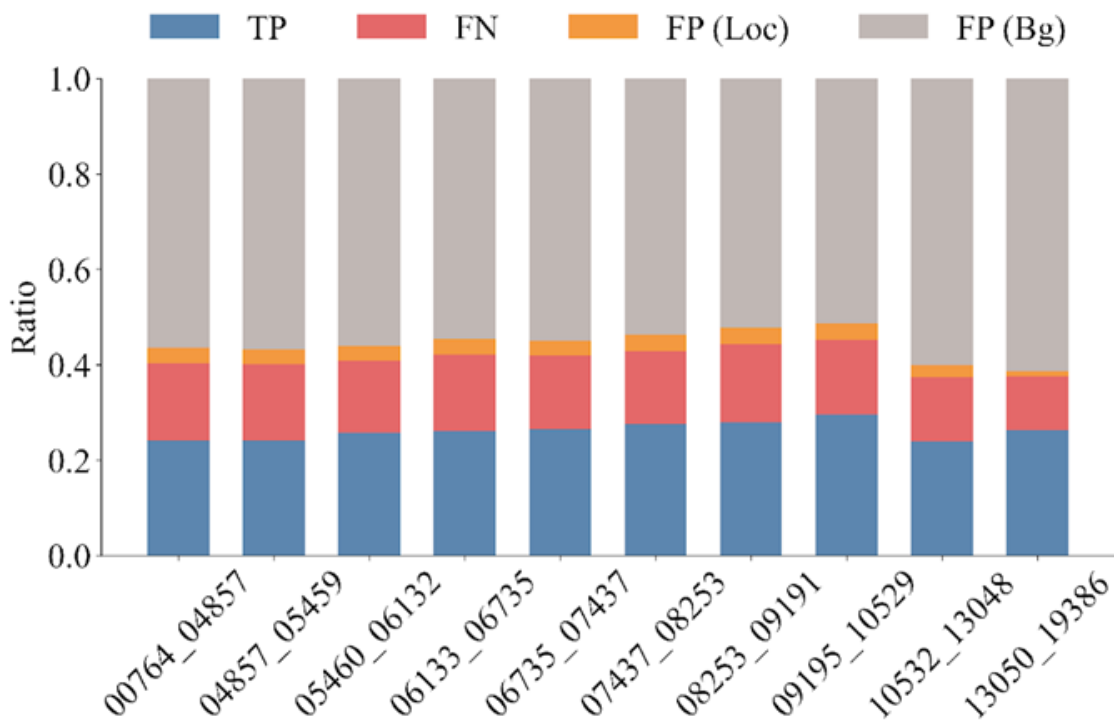
Baseline Performance: The Impact of Snow



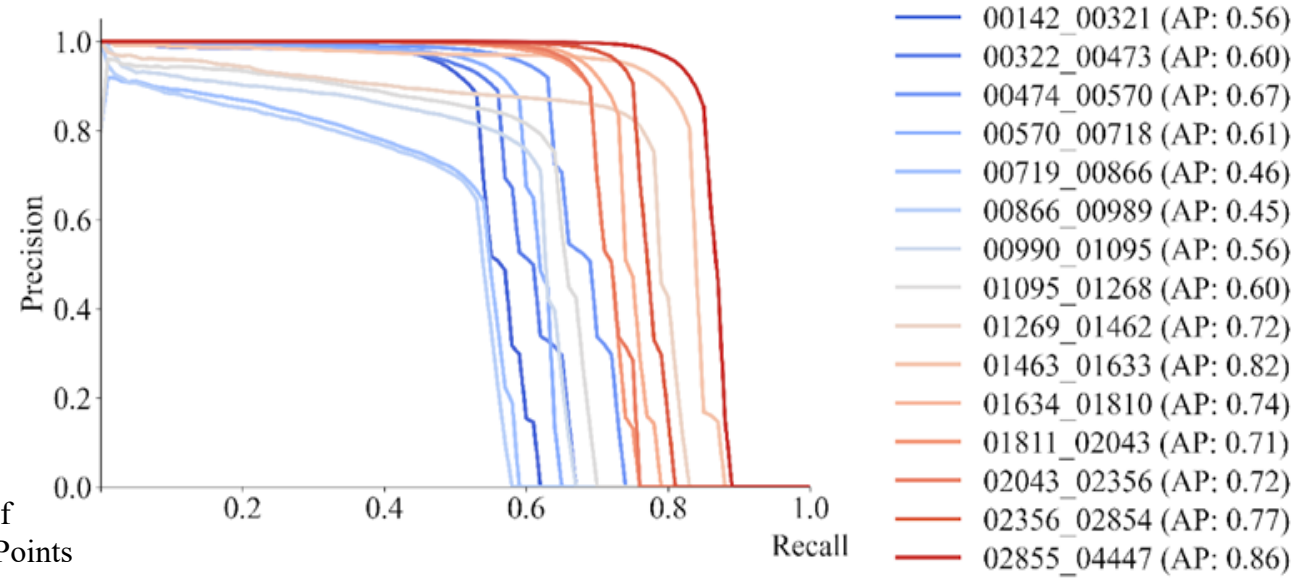
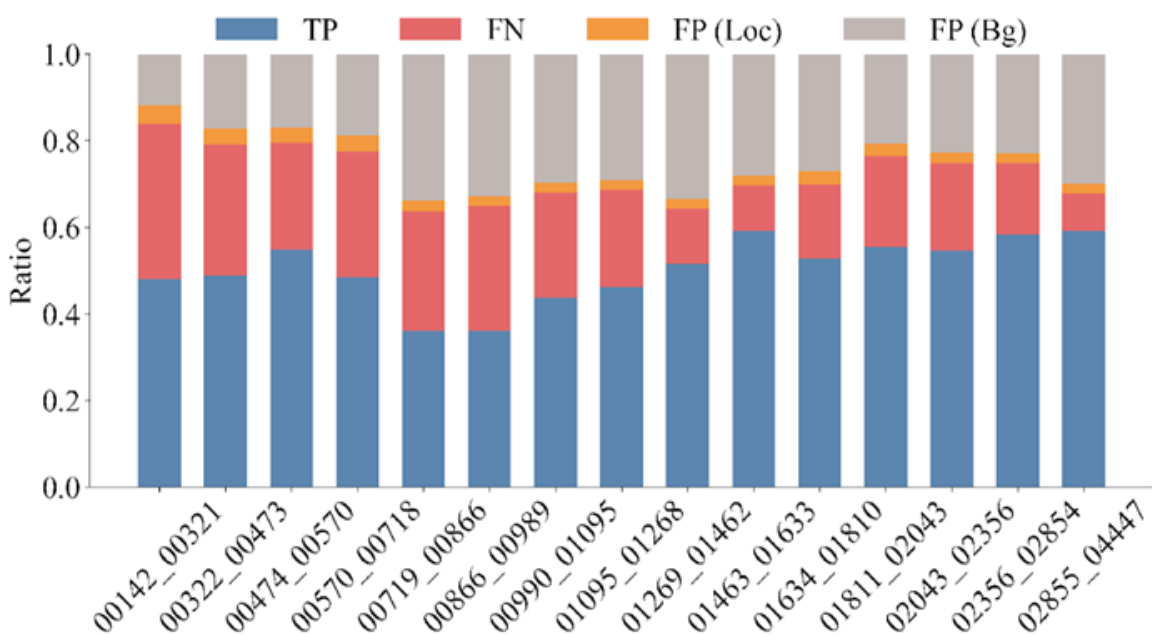
Ablation Study: Impact of Falling Snow



Error Matrix and Recall Analysis - WADS-3D



Error Matrix and Recall Analysis - CADC



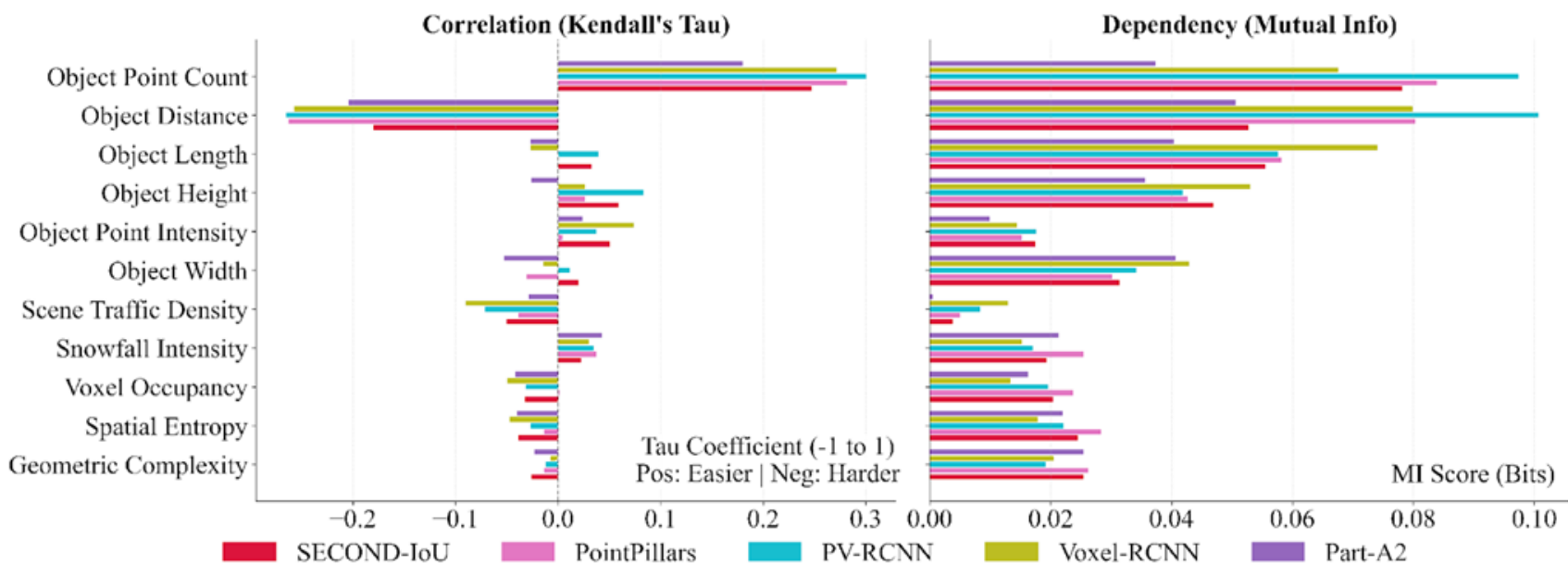
Meta-info Analysis

- Kendall Tau correlation
- Mutual Information (dependencies)
- Multivariate importance (SHAP)

Meta-feature	Object-level	Meta-feature	Scene-level
1	Object Point Count	7	Traffic Density
2	Object Distance	8	Snowfall Intensity
3	Object Length	9	Spatial Entropy
4	Object Width	10	Voxel Occupancy
5	Object Height	11	Geometric Complexity
6	Object Point Intensity		

Meta-analysis on recall – WADS-3D

- Object point count
- Object distance

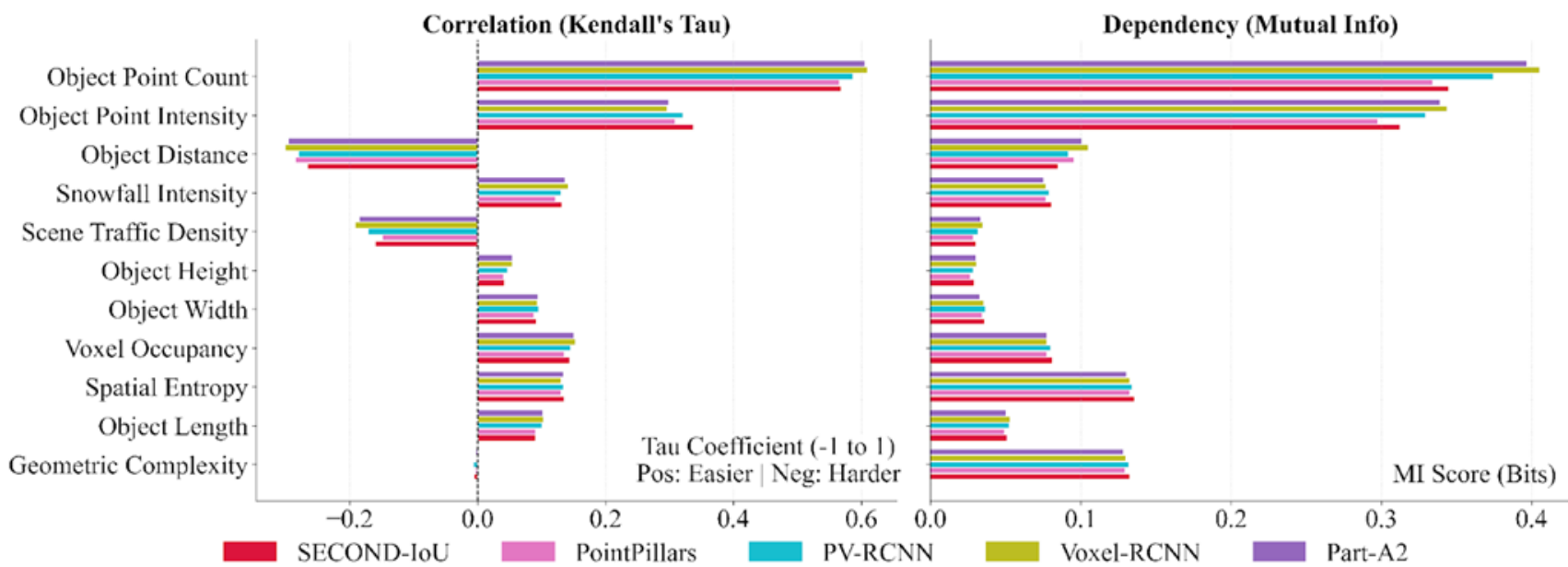


Feature Importance (SHAP)					Relative Importance
0.33	0.33	0.28	0.27	0.21	
0.15	0.24	0.25	0.22	0.24	
0.12	0.12	0.11	0.13	0.10	
0.13	0.10	0.11	0.10	0.13	
0.05	0.05	0.03	0.06	0.09	
0.05	0.05	0.04	0.04	0.08	
0.05	0.02	0.07	0.07	0.04	
0.03	0.04	0.03	0.03	0.03	
0.03	0.02	0.03	0.03	0.03	
0.04	0.01	0.02	0.03	0.02	
0.03	0.02	0.02	0.01	0.02	
SECOND (IoU)	Point Pillars	PV RCNN	Voxel RCNN	Part-A2	



Meta-analysis on recall - CADC

- Object point count
- Object point intensity



Feature Importance (SHAP)					Relative Importance
0.54	0.54	0.55	0.57	0.56	
0.19	0.18	0.19	0.17	0.17	
0.05	0.07	0.05	0.06	0.05	
0.05	0.05	0.04	0.04	0.04	
0.04	0.03	0.05	0.05	0.05	
0.03	0.04	0.03	0.03	0.03	
0.02	0.02	0.02	0.02	0.02	
0.02	0.02	0.02	0.02	0.02	
0.02	0.02	0.02	0.02	0.02	
0.01	0.01	0.02	0.01	0.01	
0.01	0.01	0.01	0.01	0.01	
SECOND (IoU)	Point Pillars	PV RCNN	Voxel RCNN	Part-A2	

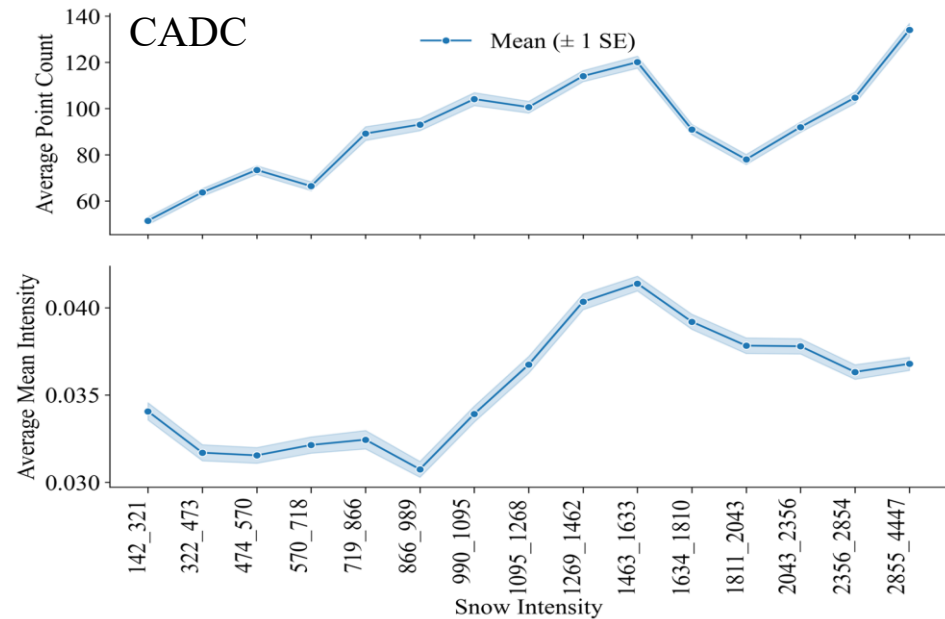
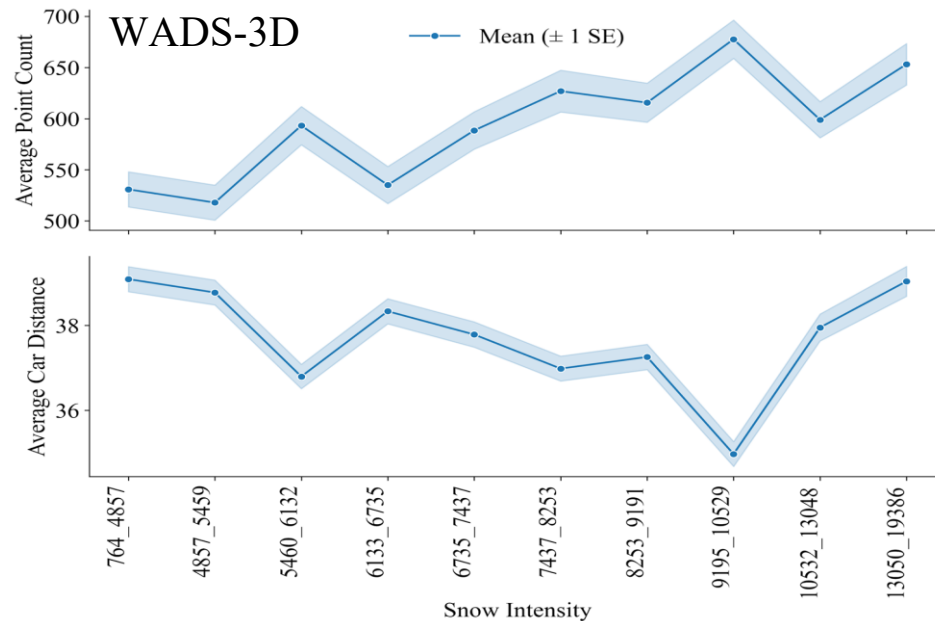


Identifying Key Meta Features

Dataset	Feature	Pearson r	p -value	Sig.
WADS-3D	Object Point Count	0.820	3.70×10^{-3}	**
	Object Distance	-0.282	0.430	NS
CADC	Object Point Count	0.733	1.86×10^{-3}	**
	Object Mean Intensity	0.694	4.12×10^{-3}	**

Bold values indicate statistical significance ($p < 0.05$).

Sig. levels: ** ($p < 0.01$), * ($p < 0.05$), NS (not significant).



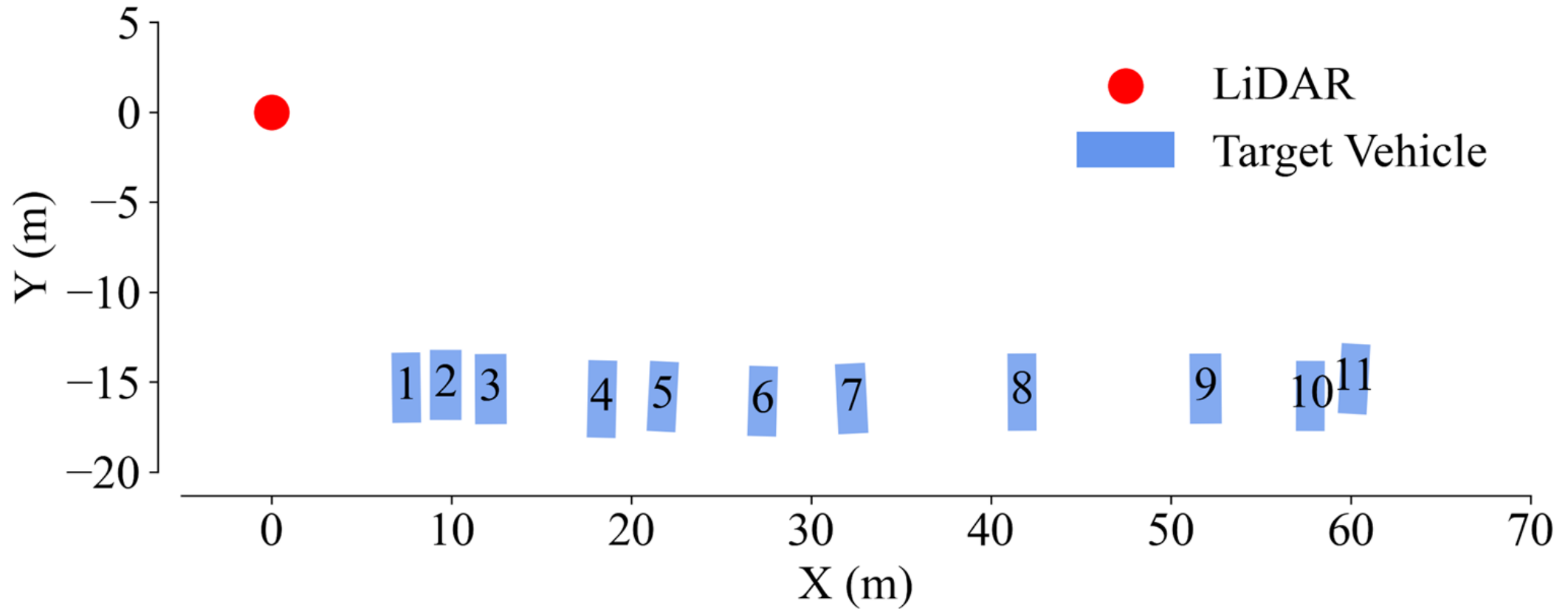
Hypothesis

- Hypothesis 1: Snow as a Scene Smoother.
- Hypothesis 2: Snow as a Geometric Enhancer.
- Hypothesis 3: Surface Reflectance Transformation.

Testing the Hypothesis - Controlled Experiment



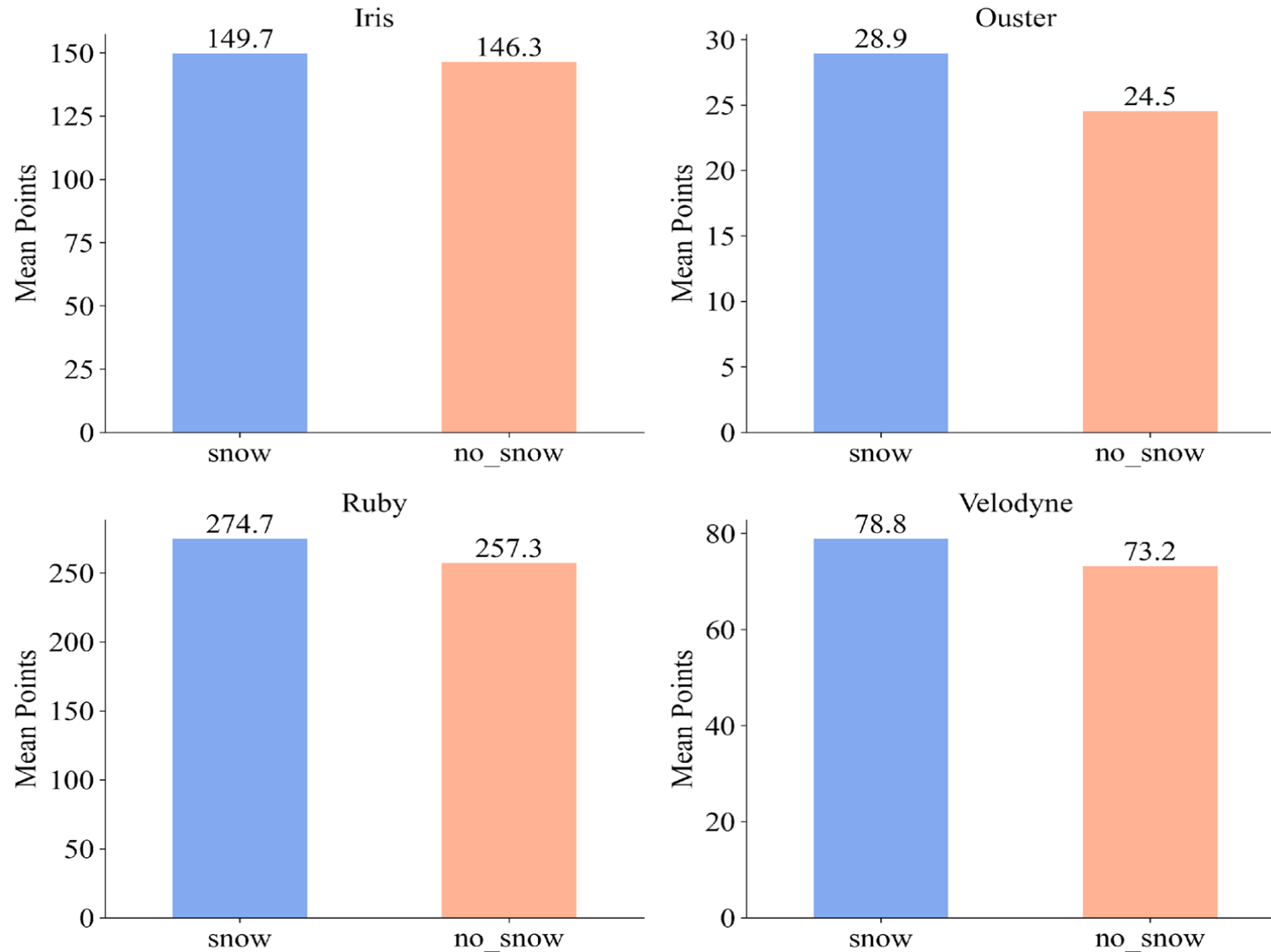
Testing the Hypothesis - Experiment Setup



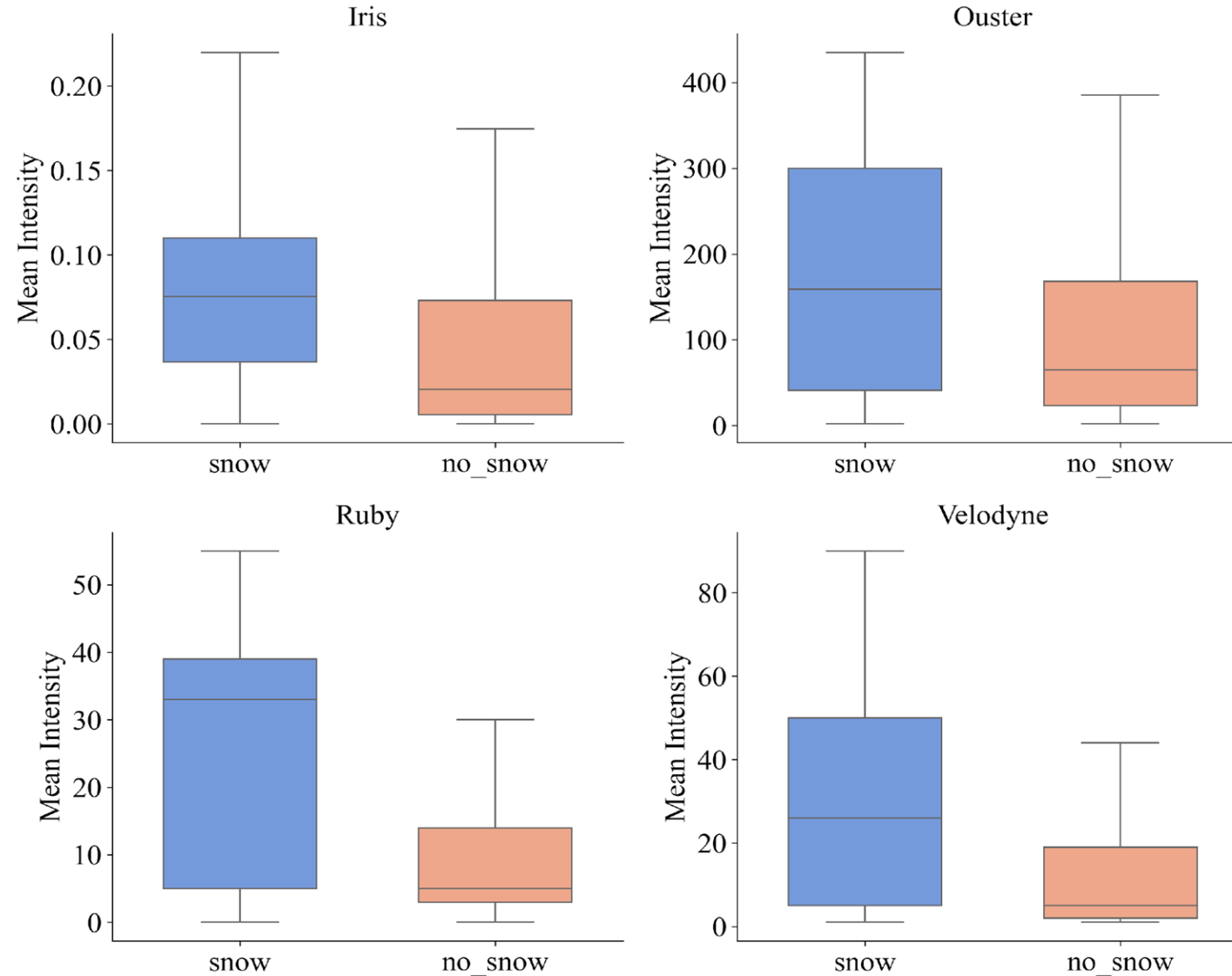
Testing the Hypothesis - Controlled Experiment

Parameter	Luminar Iris (1550nm Raster)	Robosense Ruby (128-Channel)	Ouster OS1-64 (850nm Gen1)	Velodyne VLP-32C (32-Channel)
Scanning Tech.	Raster Scan (MEMS)	Mechanical Spinner	Mechanical Spinner	Mechanical Spinner
Wavelength	1550 nm	905 nm	855 nm	903 nm
Channels	1 (Equivalent to 640 lines)	128	64	32
Vertical FOV	26° (0° to 26°)	40° (-25° to +15°)	33.2° (±16.6°)	40° (-25° to +15°)
Vertical Resolution	0.05° (Uniform)	0.1° (Non-uniform)	0.52° (Uniform)	0.33° (Non-uniform)
Horizontal FOV	120°	360°	360°	360°
Horizontal Resolution	0.05°	0.2°/0.4°	0.17° (at 1024 mode)	0.1° – 0.4°
Rotation / Scan Rate	1 – 30 fps	10 Hz / 20 Hz	10 Hz / 20 Hz	5 – 20 Hz
Raw Intensity	7-bit (0–127)	8-bit (0–255)	16-bit (0–65535)	8-bit (0–255)

Analysis 1: Object Point Count Inflation

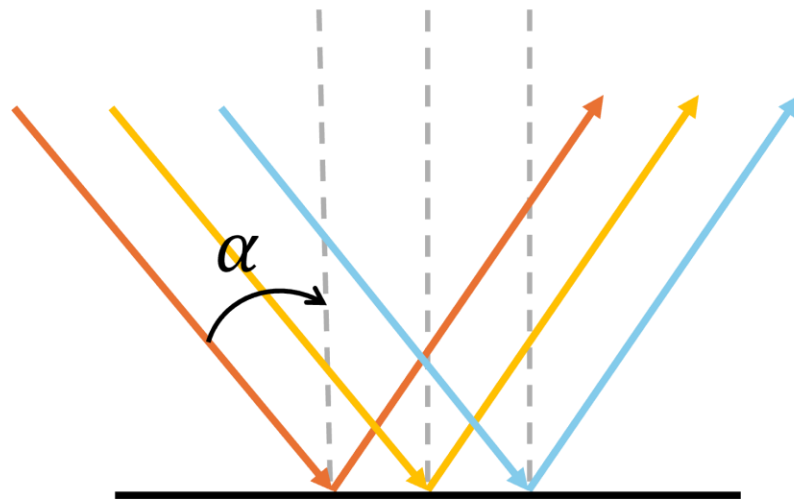


Analysis 2: Object Intensity Inflation

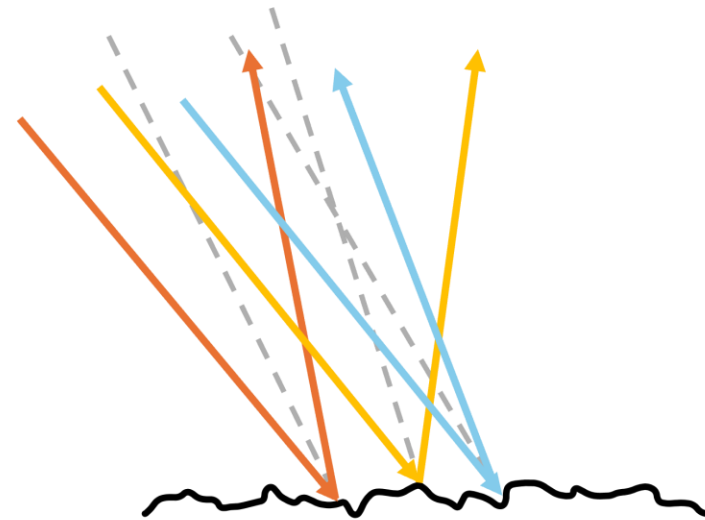


Analysis 3: Weak return vs Angle of incidence

- Weak returns: the first quartile (Q1) of intensity data
- Angle of incidence: the angle between the surface normal and the incoming laser beam

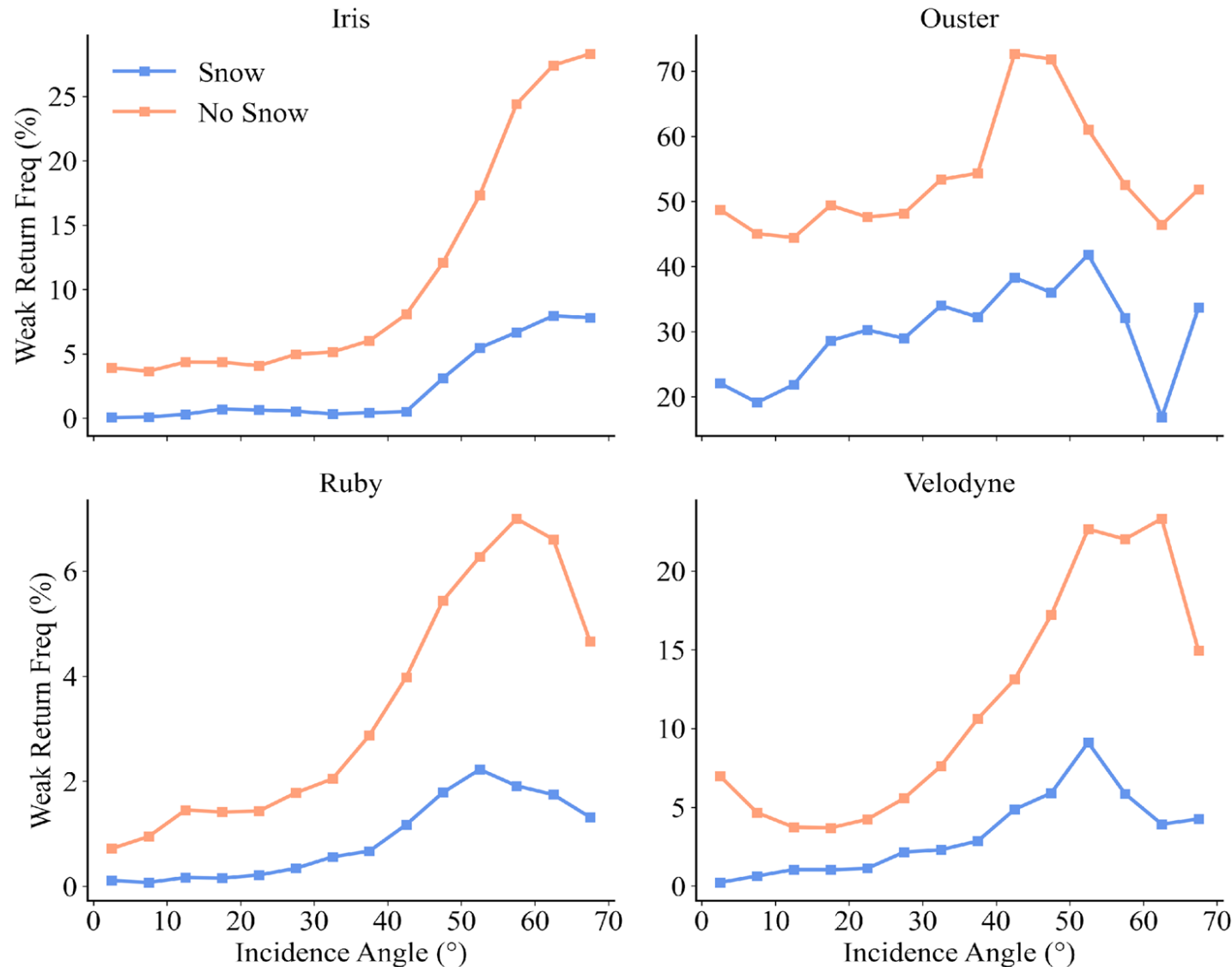


Specular Reflection



Diffuse Reflection

Analysis 3: Weak return vs Angle of incidence



Conclusion

- Falling snow acts as noise
- Accumulated snow changed object characteristics
 - Increasing point count
 - Higher point intensity
 - Mitigating weak returns

Future Work

- Apply neural detector on the controlled experiment
- Snow simulator for both falling and accumulated snow

Q & A

